

Face-2-Face with Employer

“As the industry specialises, the demand for domain experts is going to increase. Hence the pay for such people should also increase. So, engineers need not abandon their love for technology to earn more. One should focus in growing in the direction of expertise,”

Rahul Hakho
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his career in 1985 as a test engineer with the then only Indian semiconductor firm, Semiconductors complex (est. 1982), Chandigarh. Today he is the designate group manager, computer communication infrastructure at ST Microelectronics, one of the leading semiconductor makers in the country. He gives you his perspective as a representative of an employer.

☐ What kind of professionals do you have/hire in this particular area:

- At entry level
- At middle level

We largely hire fresh graduates. It makes sense as we need professionals trained according to the needs of our industry. And hence to put them in sync with the industry we give them the initial training. No Indian institute specifically trains for semiconductors. That doesn't mean the institutes aren't good. Rather they train them to fit in any engineering environment. As far as we are concerned, we pick up largely electronic and communication engineers.

Since the industry is so specialised, it is extremely rare that we pick up anyone from outside the industry at the mid-level.

☐ What is a typical career path(s) in your organization?

Typically one is hired as a design engineer and then the person works and learns at this level for 2-3 years; after this, one can become a senior design engineer. The distinction being in terms of responsibility and of course pay. One can reach up to the level of a team leader after about 10 years.

☐ Are these career paths fairly representative? Are there any other major alternative paths in other kind of organizations?

The whole industry is a knowledge-intensive industry. Hence the career path is pretty much the same in the entire industry. No one moves up without learning and executing substantially. Though after 4-5 years, when someone does learn the craft of chip-making, one can switch to the EDA tool-making industry. Making an EDA tool is even more specialised than chip making as they have to make very precise instruments. These instruments are in turn used to make chips. Hence it helps to have people who have made chips on-board.

☐ How does compensation and responsibility typically grow over time?

We design chips for a number of industries, such as auto, telecom, medicine, and for each of the chips that are underway, there are various teams working on them. The design engineer is a small but significant part of several such small teams. Overtime, he gains more responsibility within such teams and in 9-10 years time he leads his own team of design engineers. At that level you have 6-7 designers under you and your salary has gone up from 4-5 lakh at the beginning to 15-16 lakh.

☐ What kind of skills and aptitude do you look for / consider important to succeed in this career?

The engineer should basically understand semiconductors and should have a good understanding of physics, mathematics and electronics. The knowledge of computer languages is not really stressed. Apart from this, the professional should be willing to learn constantly, as learning and its application will be tested constantly.